

3. Methodology

3.1 Molecular Representation and Input Modalities

The proposed framework, **MoLFormer-Multimodal v3**, integrates four complementary molecular representations to capture structural, topological, and physicochemical information simultaneously. Given a molecule, inputs are prepared as: (i) a SMILES string tokenised for a pre-trained transformer encoder; (ii) an extended-connectivity fingerprint (ECFP) of length F_1 ; (iii) a MACCS key fingerprint of length F_2 ; and (iv) a vector of F_3 RDKit physicochemical descriptors. These modalities encode orthogonal information — the SMILES encoder captures sequential context and aromaticity, ECFP encodes circular substructure neighbourhoods, MACCS encodes predefined pharmacophoric patterns, and descriptors encode global properties such as logP and molecular weight.

3.2 Contextual SMILES Encoding via Multi-Layer Pooling

The SMILES sequence is encoded by MoLFormer-XL (IBM), a 12-layer linear-attention transformer pre-trained on 1.1 billion SMILES strings. To exploit the representational hierarchy of deep transformers, we aggregate the hidden states of the top K encoder layers via a learned weighted sum rather than relying solely on the final layer:

$$\mathbf{H} = \sum_{k=1}^K \alpha_k \mathbf{H}^{(L-K+k)}, \quad \boldsymbol{\alpha} = \text{softmax}(\mathbf{w}), \quad \mathbf{w} \in \mathbb{R}^K$$

where $\mathbf{H} \in \mathbb{R}^{B \times L \times D}$ is the blended hidden state matrix ($D = 768$), L is the padded sequence length, and B is the batch size. The scalar weights $\mathbf{w}$ are learned jointly during fine-tuning (default $K = 4$, selecting layers 9-12).

To obtain a fixed-length molecular representation from the variable-length token matrix, three pooling strategies are applied and averaged:

$$\mathbf{h}_{\text{CLS}} = \mathbf{H}_{:,0,:}$$

$$\mathbf{h}_{\text{mean}} = \frac{\sum_{i=1}^L m_i \mathbf{H}_{:,i,:}}{\sum_{i=1}^L m_i}, \quad m_i \in \{0, 1\}$$

$$\mathbf{h}_{\text{attn}} = \sum_{i=1}^L \beta_i \mathbf{H}_{:,i,:}, \quad \boldsymbol{\beta} = \text{softmax}(\tilde{\mathbf{s}})$$

$$\tilde{s}_i = \begin{cases} \mathbf{w}_a^\top \mathbf{H}_{:,i,:} & m_i = 1 \\ -\infty & m_i = 0 \end{cases}$$

where m_i is the attention mask (0 for padding tokens) and $\mathbf{w}_a \in \mathbb{R}^D$ is the learnable scoring vector. The final SMILES embedding is:

$$\mathbf{s} = \frac{\mathbf{h}_{\text{CLS}} + \mathbf{h}_{\text{mean}} + \mathbf{h}_{\text{attn}}}{3} \in \mathbb{R}^{B \times D}$$

This ensemble mitigates the known biases of individual pooling strategies: CLS tokens may underfit for small molecules, mean pooling dilutes sparse functional-group signals, and attention pooling alone is prone to collapse onto a single token.

3.3 Modality Projection

Each non-SMILES input is projected to the shared embedding space $\mathbb{R}^D$ via a two-stage feedforward network with GELU activations and LayerNorm applied after each linear stage:

$$\mathbf{e}_m = \text{LN}\left(\text{GELU}\left(\mathbf{W}_m^{(2)} \cdot \text{LN}\left(\text{GELU}(\mathbf{W}_m^{(1)} \mathbf{x}_m + \mathbf{b}_m^{(1)})\right) + \mathbf{b}_m^{(2)}\right)\right), \quad m \in \{\text{ECFP}, \text{MACCS}, \text{DESC}\}$$

Placing LayerNorm after each GELU stage (rather than before) ensures every projected token enters the cross-modal transformer with near-unit variance, maintaining scale consistency with the normalised SMILES embedding $\mathbf{s}$.

3.4 Modality-Aware Cross-Modal Transformer

The four modality embeddings $\{\mathbf{s}, \mathbf{e}_{\text{ECFP}}, \mathbf{e}_{\text{MACCS}}, \mathbf{e}_{\text{DESC}}\}$ are assembled into a token sequence $\mathbf{T} \in \mathbb{R}^{B \times 4 \times D}$. Without positional or type signals, a standard transformer cannot distinguish which token corresponds to which modality. We therefore add learnable modality type embeddings:

$$\mathbf{T}' = \mathbf{T} + \mathbf{E}_{\text{type}}, \quad \mathbf{E}_{\text{type}} \in \mathbb{R}^{4 \times D}$$

where $\mathbf{E}_{\text{type}}$ is initialised with Xavier uniform weights to match the scale of the LayerNorm-normalised projections. The augmented sequence is processed by an N_c -layer Pre-LayerNorm transformer encoder (default $N_c = 3$):

$$\mathbf{T}'' = \text{CrossModalTransformer}(\mathbf{T}') \in \mathbb{R}^{B \times 4 \times D}$$

Pre-LN placement stabilises gradient flow during fine-tuning of a partially-frozen backbone.

3.5 Gated Fusion and Prediction

A per-modality, per-dimension gate is applied to selectively suppress or amplify each modality's contribution before fusion:

$$\mathbf{G} = \sigma(\text{LN}(\mathbf{W}_g \mathbf{T}'' + \mathbf{b}_g)) \in \mathbb{R}^{B \times 4 \times D}$$

LayerNorm before the sigmoid prevents saturation from large pre-activation magnitudes. The gated tokens are flattened and projected:

$$\mathbf{f} = \mathbf{W}_{\text{fuse}} \cdot \text{vec}(\mathbf{G} \odot \mathbf{T}'') \in \mathbb{R}^{B \times D}, \quad \mathbf{W}_{\text{fuse}} \in \mathbb{R}^{D \times 4D}$$

The fused representation $\mathbf{f}$ is passed to a halving MLP classifier of depth N_d with GELU activations, LayerNorm, and progressive dropout, yielding a scalar prediction $\hat{y} \in \mathbb{R}$.

3.6 Regularisation and Transfer Learning

To prevent catastrophic forgetting of chemical pretraining, the bottom N_f encoder layers of MoLFormer-XL are frozen at initialisation (default $N_f = 4$). The remaining layers are fine-tuned using layer-wise learning rate decay (LLRD), assigning higher learning rates to upper layers. Dropout is applied throughout (rate p , with $p/2$ at the final classifier layer), and all non-BERT parameters are regularised via weight decay.